

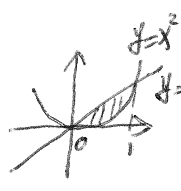
Math 241, F14  
Quiz 10

Name: Answer  
SSN: xxx-xx-

**Instructions:** There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

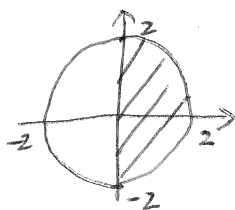
**Problem 1.** (6 points) Calculate the following integrals:

$$\begin{aligned} \text{(a)} \quad & \int_0^1 \int_1^2 (4x^3 - 9x^2y^2) dy dx \\ &= \int_0^1 [4x^3y - 3x^2y^3]_{y=1}^{y=2} dx \\ &= \int_0^1 [8x^3 - 24x^2] - [4x^3 - 3x^2] dx \\ &= \int_0^1 4x^3 - 21x^2 dx \\ &= [x^4 - 7x^3]_{x=0}^{x=1} \\ &= [1 - 7] - [0] \\ &= -6 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad & \iint_D x + 2y dA, \text{ where } D \text{ is the region} \\ & \text{enclosed by } y = x \text{ and } y = x^2. \\ & \begin{aligned} & \begin{cases} y = x \\ y = x^2 \end{cases} \Rightarrow x^2 - x = 0 \\ & x(x-1) = 0 \Rightarrow x = 0, x = 1 \end{aligned} \\ & \Rightarrow D = \{(x, y) \mid 0 \leq x \leq 1, x^2 \leq y \leq x\} \\ & \iint_D x + 2y dA = \int_0^1 \int_{x^2}^x x + 2y dy dx \\ &= \int_0^1 [xy + y^2]_{y=x^2}^{y=x} dx = \int_0^1 [(x^2 + x^2) - (x^3 + x^4)] dx \\ &= \int_0^1 2x^2 - x^3 - x^4 dx \\ &= \left[ \frac{2}{3}x^3 - \frac{1}{4}x^4 - \frac{1}{5}x^5 \right]_{x=0}^{x=1} \\ &= \left( \frac{2}{3} - \frac{1}{4} - \frac{1}{5} \right) - 0 = \frac{13}{60} \end{aligned}$$


**Problem 2.** (4 points) Let  $R = \{(x, y) \mid x^2 + y^2 \leq 4, x \geq 0\}$ . Compute the double integral

$\iint_R \sqrt{4 - x^2 - y^2} dA$  by changing to polar coordinates.



$$R = \{(r, \theta) \mid -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}, 0 \leq r \leq 2\}$$

$$\sqrt{4 - x^2 - y^2} = \sqrt{4 - (x^2 + y^2)} = \sqrt{4 - r^2}$$

$$\begin{aligned} \Rightarrow \iint_R \sqrt{4 - x^2 - y^2} dA &= \int_{-\pi/2}^{\pi/2} \int_0^2 \sqrt{4 - r^2} r dr d\theta \\ &= \int_{-\pi/2}^{\pi/2} \left[ -\frac{1}{3}(4 - r^2)^{3/2} \right]_{r=0}^{r=2} d\theta \\ &= \int_{-\pi/2}^{\pi/2} \left[ \left(-\frac{1}{3} \cdot 0\right) - \left(-\frac{1}{3} \cdot 8\right) \right] d\theta \\ &= \int_{-\pi/2}^{\pi/2} \frac{8}{3} d\theta = \frac{8}{3} \pi \end{aligned}$$